(a)

In [6]:
```
import numpy as np
from scipy.optimize import fsolve

def f(x):
    return x**2 - np.exp(-x) - 2 * np.sin(x)

initial_guess = 1.5
root_found = fsolve(f, initial_guess)

print("The root found is : ",root_found)
```

```
The root found is :  [1.48958864]
```

(b)

In [8]:
```
import numpy as np
from scipy.optimize import fsolve

def f(t):
    return 10 * t - 4.9 * t**2

initial_guess = 1.5
root_found = fsolve(f, initial_guess)

print("Time take projectile returns to initial height: ",root_found,"Seconds")
```

```
Time take projectile returns to initial height:  [2.04081633] Seconds
```

(c)

In [11]:
```
import numpy as np
from scipy.optimize import fsolve

def f(x):
    return x**2 - 3

initial_guess = 1.5
root_found = fsolve(f, initial_guess)

print("The Square root of 3 is: ",root_found)
```

```
The Square root of 3 is:  [1.73205081]
```

(d)

In [14]:
```python
import numpy as np
from scipy.optimize import fsolve

def f(L):
    return L - (100*9.8)/(4*np.pi*np.pi)

initial_guess = 1.5
root_found = fsolve(f, initial_guess)

print("Length of the pendulum for T=10 : ",root_found,"meters")
```

```
Length of the pendulum for T=10 :  [24.82368999] meters
```

(e)

In [15]:
```python
import numpy as np
from scipy.optimize import fsolve

def f(t):
    return 5 * (1 - np.exp(-t / 10)) - 2.5

initial_guess = 1.5
root_found = fsolve(f, initial_guess)

print("Voltage across the capacitor : ",root_found,"V")
```

```
Voltage across the capacitor :  [6.93147181] V
```

In [ ]:
```python

```